

Experiment 'p_1_4_account_balance' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_4_account_balance

Natural language statement: Write a method that returns the balance of an account after the first, second, and third year. The account has an initial balance of \$1,000 and earns 5 percent interest per year.

Method signature: p_1_4_account_balance() returns (first: real, second: real, third: real)

Ensures

- `first == 1000.0 + 1000.0 * 0.05`
- `second == first + first * 0.05`
- `third == second + second * 0.05`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the balance of an account after the first, second, and third
→ year. The account has an initial balance of \$1,000 and earns 5 percent interest
→ per year.

The signature should be:

`p_1_4_account_balance()` returns (first: real, second: real, third: real)

The method should respect the following contract:

ensures first == 1000.0 + 1000.0 * 0.05, ensures second == first + first * 0.05, ensures
→ third == second + second * 0.05

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLZLaZ4Zq71WVGy1vZQ7Yu1LI5g

Model: gpt-5.5-2026-04-23

Created at: 1780483365

Finish reason: stop

Usage: 390 tokens in, and 184 tokens out

```
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Final Program

```
method p_1_4_account_balance() returns (first: real, second: real, third: real)
  ensures first == 1000.0 + 1000.0 * 0.05
  ensures second == first + first * 0.05
  ensures third == second + second * 0.05
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

}

Total Token Usage

Input tokens: 390

Output tokens: 184

Reasoning tokens: 117

Sum of 'total tokens': 574

Experiment Timings

Overall Experiment started at 1780483365161, ended at 1780483373201, lasting 8040ms (8.04 seconds)

Iteration #1 started at 1780483365163, ended at 1780483373201, lasting 8038ms (8.04 seconds)